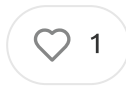


FREE

Updates to the Ultimate Vitamin K2 Resource



CHRIS MASTERJOHN, PHD
DEC 29, 2016



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In response to reader feedback and my own experience using [The Ultimate Vitamin K₂ Resource](#), I made several updates, including making the second infographic [easier to read](#), [improvements to the performance of the page](#), [additions](#) and [improvements](#) to the searchable database, and [supplements](#) added to the review. At the request of a reader, I also made [a really easy way to support this site with Amazon purchases](#). As I intend this to be a continually updated perpetually useful resource, consider this the first round of updates.

Improved Clarity to the Second Infographic

Several people found the third row of the [second infographic](#), “Why It Matters What Type of Vitamin K We Eat,” difficult to read. We changed the font from all-caps to regular capitalization and made the background darker, and it is now much easier to read.

The Site Now Loads Three Times Faster

My site has, since it started, loaded rather slowly, and that's because I had a budget of near zero when I started it and got a deal of one year free hosting. Being free, it was nothing to complain about. However, The Ultimate Vitamin K2 Resource has a lot to load and is generating tons more traffic to the site, which means its time for an oil change. I migrated my site to [WP Engine](#), which instantaneously cut the load times on my pages by 65%. WP Engine offers a lot of assistance in further optimizing your site for performance and speed. I have yet to take advantage of any of that, but hope to in January. Nevertheless, simply moving from one server to another has caused everything to operate at three times the speed.

My choice to move to [WP Engine](#) happened *after* a failed migration to another company's servers. I'll keep them nameless for the time being, but my own hastiness combined with their inadequate technical assistance and customer service caused my site to temporarily crash worldwide. I apologize to any of you who experienced any frustration as a result of this during the transition. The WP Engine migration was smooth as a baby's bottom for two reasons: first, most of it is automated with a fantastic plugin; second, their customer service is off-the-charts, with unlimited continuous access to both phone and live chat. While the downtime the site experienced is embarrassing to me, I'm incredibly happy with the end result. It's

pricey, but I intend this site to be awesome in every way in 2017, and awesome content is only as awesome as the speed and performance of the site that hosts it.

The Vitamin K2 Searchable Food Database Now Opens in a New Tab

Originally, the [searchable database](#) opened in the same tab. If you wanted to do more than one search, you had to hit the back button, which brought you all the way to the top of the article. All this was far more terrible than it sounds, because the slow site speed prior to the WP Engine migration was causing each of these events to take 10 seconds! Imagine doing five keyword searches, with each search taking 10 seconds, each time going back to the page taking 10 seconds, and then going back to the database with several more seconds, depending on your scrolling skills. A nightmare.

Now, the database opens in a new tab, so you are never more than a tab-switch away from it for a new search. Plus, the site migration has made each keyword search load three times faster. Together, searching the database now offers a much better experience.

Emu Oil, Green Pastures Butter Oil, and Over a Dozen Other Foods Have Been Added to the Database

Many people requested data on Walkabout Australian Emu Oil and Green Pastures Butter Oil. These have now been added to the database. The data for these products comes from a set of foods sent by the [Weston A. Price Foundation](#) to [VitaK](#) for independent analysis. Since the foundation has made the original lab report publicly available, I was able to add these foods to the database and be confident that I was maintaining the same level of rigor as before. To date, every single entry in the database has the exact source of information available to the user simply by clicking on “view more details” under the specific result, and the sources come either from the peer-reviewed scientific literature or independent analyses documented with original lab reports.

Other foods added to the database include oysters, fish roe, shrimp, conventional and pastured chicken liver, cheeses and butters of different varieties with name brands or local farm sources listed, egg yolks, tallow, duck fat, lard, and cod liver oil. If you're salivating already, go on and [search your heart out](#) in the database.

10 Supplements Have Been Added to the Review

In response to audience requests, the following supplements have been added to the [comprehensive review](#). I have not changed my top three recommendations in the main text. These include [mixed MK-4 and MK-7](#) supplements, [natto-derived MK-7](#)

supplements, and an apparently [synthetic MK-7 supplement](#). Additionally, I created two new categories: [High-Dose MK-4](#), and [Food-Based Supplements](#).

Mixed MK-4 and MK-7 Supplements

[Pure Encapsulations Synergy K](#) — 1,000 IU Vitamin D₃, 1 mg MK-4, 500 µg K₁, 45 µg MK-7. 47 cents per capsule, 9 cents per day to average 200 µg/d. \$56 on [Amazon](#), where it is fulfilled by Amazon and eligible for Prime. Free of GMOs and allergens. Taken to yield an average close to 200 µg K₂, the amounts of vitamin D (200 IU) and MK-7 (9 µg) are rather low compared to the [Innovix Full-Spectrum K₂](#) (20 µg MK-7) or the [Thorne Research D/K₂](#) (1,000 IU vitamin D), but the unique combination of the three vitamins may be optimal for some people whose nutritional needs fit it just right.

Note: This is slightly and almost imperceptibly more expensive than the Innovix and Country Life supplements, so it is listed in the comprehensive review right underneath them as the last one in the “Mixed MK-4 and MK-7 Supplements” section. Since price is the organizing principle in each section, I added the following statement to the end of the Country Life entry for clarity: “The lack of transparency about the composition of this supplement is frustrating and I don't recommend it.” For mixed MK-4/MK-7 supplements, I recommend choosing between Innovix and and Pure Encapsulations based on your nutritional needs. I think the mix in Innovix is more suitable to a larger number of people so it remains my top pick among the MK-4/MK-7 combinations.

[Maxx Labs Vitamin K₂ Complex](#) — 500 µg MK-4, 100 µg MK-7, 100 mg calcium from calcium citrate. 20 cents per capsule, 7 cents per day to average 200 µg/d. \$17.98 on [Amazon](#), where it is fulfilled by Amazon and eligible for Prime. Be careful to avoid the subscription setting if you only want to order one bottle. Free of GMOs and allergens.

Note: This is less expensive than the otherwise similar [Innovix Full-Spectrum K₂](#) (7 vs. 9 cents per day) and is therefore placed second in the “Mixed MK-4 and MK-7 Supplements” section. At the present time, I find the sales and marketing associated with the Innovix supplement more accurate and I am maintaining them as my top recommendation for that reason. My reasons for not giving that spot to [Life Extension Super K](#) is the high concentration of K₁, and this is already clear in the review.

As examples of the the inaccurate sales and marketing around the Maxx Labs product, the label itself misspells “menatetrenone” and it claims that osteocalcin is “needed to bind calcium into the matrix of bone” and “also appears to prevent calcium from depositing into arteries.” As described in the clickable “Biochemistry” section underneath “The Health Benefits of Vitamin K₂” in the [resource](#), osteocalcin is a hormone that is released from bone to regulate insulin, testosterone, and energy metabolism, and the control of calcium distribution is mediated by MGP and possibly an assortment of other lesser known vitamin K-dependent proteins. These errors are actually disturbingly common in the marketing materials of supplement manufacturers, but they are more prominent on the label in the Maxx Labs product, and I thought about this product more intensively than others when trying to decide whether to bump Innovix out of my top three list.

Vin Kutty explained in a [blog post](#) why he developed the Innovix supplement, and while I don't 100% agree with everything he says, his high level of accuracy, apparent

authenticity, and attention to detail make me more confident in the [Innovix](#) supplement. So, I went with trust over price in this case.

Natural MK-7 From Natto

[Natural Factors K₂](#) — 100 µg MK-7. 21 cents per capsule, 42 cents per day to achieve 200 µg. \$12.57 from [Amazon](#). Derived from non-GMO natto. The company has a unique farm-to-capsule model where it controls everything that goes into its supplements from the soil to the encapsulation.

Note: As a result of its price, it ranks third in the “Natural MK-7 From Natto” section. When all of the natural MK-7-only supplements are pooled together, this supplement ranks fifth for price. If you don't mind paying extra, it is worth researching the company's unique commitment to vertical integration and sustainability to see if the philosophical approach of the company is something you would want to support with the premium you have to pay for this supplement.

[Natural Factors D₃ & K₂](#) — 1000 IU vitamin D, 120 µg MK-7. 35 cents per softgel, 58 cents per day taken to average 200 µg/d, or 70 cents per day to yield 240 µg. Identical in price (\$20.97) between [Amazon](#) and [iHerb](#). MK-7 is derived from non-GMO natto. The company has an unusual farm-to-capsule model where it controls everything that goes into its supplements from the soil to the encapsulation. Taken to yield an average daily dose of 200 µg K₂/d, it yields 1,667 IU vitamin D; taken twice a day to yield 240 µg K₂, it yields 2000 IU of vitamin D, both of which are more than the 1,000 IU of vitamin D in [Thorne Research D/K₂](#). However, the Thorne product contains MK-4 and this product contains MK-7. This is the only product that combines relatively high doses of vitamin D and MK-7.

Note: Since I don't currently have a section on vitamin D combinations, this goes into the “Natural MK-7 From Natto” section, where it places eighth for price. When the natural MK-7 supplements are pooled, it places eleventh for price. If you are concerned about cost-effectiveness, you could easily replicate this combination with much less expensive MK-7 and vitamin D supplements. However, my comments above about the company's philosophical approach apply here as well.

Synthetic MK-7 Supplements

[K-Force](#) — 5,000 IU vitamin D3, 180 µg MK-7. \$1.13 per capsule, and \$1.13 per day to yield 180 µg or \$1.20 per day to yield an average of 200 µg/d. \$67.95 on [Amazon](#), where it can be obtained with free shipping but is not fulfilled by Amazon or eligible for Prime. The company lists the MK-7 as soy-free but does not clarify its origin. Presumably it is synthetic, but it may be derived from fermented chickpeas. This supplement is useful specifically to people who need to take 5,000 IU of vitamin D per day.

Note: I believe that 5,000 IU of vitamin D is more vitamin D than most people need on a daily basis, so I do not recommend this as a general-purpose MK-7 supplement, but I believe that the MK-7 would help make the vitamin D safer and think it is better to take something like this than to take 5,000 IU of vitamin D on its own. Nevertheless, it

would be much more cost-effective to make your own combination from less expensive vitamin D and MK-7 supplements.

High-Dose MK-4

Supplements where the lowest dose contains 5 mg or more of MK-4 are included in this section. It is impractical to use these supplements to reach an average dose of 200 µg/d. They are primarily useful as a means of reaching the pharmacological dose of 45 mg/d that has been used to treat osteoporosis, to prevent the occurrence of hepatocellular carcinoma in women with viral cirrhosis, and to prevent the recurrence of the same disease in people who have already been treated for it. Although high-dose MK-4 supplements are available over-the-counter and have a low risk of side effects, these treatments are pharmacological rather than nutritional in nature. Therefore, I recommend using them under the supervision of the physician who is overseeing treatment for one of these conditions.

[Advanced Orthomolecular Research Peak K2](#) — 15 mg MK-4. 35 cents per capsule and \$1.04 per day to reach 45 mg. Less expensive on [Amazon](#) (\$31.30) than [iHerb](#) (\$34.56). On Amazon, fulfilled by Amazon and eligible for Prime. Free of common allergens.

Note: This company also makes a [low-dose supplement](#), but it can't be ordered online and the company is not transparent about its price, so I am not including it in this review.

[Relentless Improvement Vitamin K2](#) — 15 mg MK-4, 60 µg MK-7. 39 cents per capsule and \$1.17 per day to reach 45 mg. \$34.95 per bottle on [Amazon](#), where it is fulfilled by Amazon and eligible for Prime free one-day shipping. Taken to provide 45 mg/d MK-4, it also provides 180 µg/d MK-7. While research suggests that MK-4 and MK-7 have different tissue distributions at low doses, it is unclear whether there is any benefit to adding a low dose of MK-7 to a far higher dose of MK-4. The vitamins are synthetic and the company guarantees a low percentage of inactive *cis* isomers.

[Carlson Labs Vitamin K2](#) — 5 mg MK-4. 18 cents per capsule and \$1.66 per day to reach 45 mg. Less expensive on [Amazon](#) than on [iHerb](#). On Amazon, the 60-capsule bottle (\$10.99) and 180-capsule bottle (\$33.26) are both equivalent in price per capsule. Both are fulfilled by Amazon and eligible for Prime, and the larger one is eligible for Prime free one-day shipping. Be careful of the subscribe and save button on the smaller bottle. iHerb only sells the 60-capsule bottle (\$14.94).

Food-Based Supplements

Supplements included in this section are those that isolate a food oil itself or a major fraction thereof rather than specifically isolating one or more forms of vitamin K₂. The nutrients in these supplements are less concentrated, but they are present in a broader network of synergists. While vitamin K₂ supplements are a great way to optimize vitamin K status in someone whose diet is otherwise good, food-based supplements are likely to be better ways of compensating for an otherwise suboptimal diet.

[Walkabout Australian Emu Oil](#) — 40 cents per gram as a liquid oil, 52 cents per gram as capsules. Each capsule contains one gram of total oil and 4 µg MK-4. This product is not available on Amazon or iHerb. Its price is identical between [Radiant Life](#), where the liquid oil (\$45) and capsules (\$52) can be ordered on the same page, and Corganic, where the [oil](#) and [capsules](#) are available on separate pages. Additional shipping charges apply to both sellers. Shipping charges for RadiantLife would be \$7.95 for the liquid oil, \$9.95 for the capsules, and free for a total order over \$125. Corganic shipping charges depend on your address. Reaching 200 µg K₂/d with this product would require 48 capsules or 3.5 tablespoons per day of the liquid oil and is obviously impractical. However, five capsules per day would yield 20 µg/d; added to a nutrient-poor diet containing only 15-20 µg/d on its own, this would double a person's K₂ intake. The oil also naturally contains a blend of essential fatty acids and other fat-soluble vitamins.

[Green Pastures X-Factor Butter Oil](#) — 43 cents per capsule, with each capsule containing 0.5 grams of oil. Less expensive on [Amazon](#) (\$43.20) than on the [Green Pastures web site](#) (\$60). On Amazon, fulfilled by Amazon and eligible for Prime free one-day shipping. However, the Green Pastures web site offers liquid oils as well as capsules and offers a greater diversity of flavors than available on Amazon. Each capsule contains 0.4 µg of K₂, mostly as MK-4. Reaching 200 µg/d would require 500 capsules per day and is obviously impractical. However, Green Pastures has in the past reported an unidentified set of quinones in the oil, which could upon further testing be shown to have additional vitamin K activity. For example, the fermentation of the oil could produce tetrahydromenaquinones, which are found in high concentrations in certain cheeses but have not been measured in the butter oil. The oil also naturally contains a blend of essential fatty acids and other fat-soluble vitamins.

Supporting the Site With Amazon Purchases

A [generous reader](#) asked me within the comments of The Ultimate Vitamin K2 Resource to put my Amazon affiliate link in the top menu of my site to make it easier to support the site with Amazon purchases. It is already the case that if you use the links in the supplement reviews to purchase supplements, a portion of the purchase will support the site. However, she was asking for a simpler way to use any Amazon purchases to support the site, regardless of whether they are linked to from the site in product reviews.

I want to maintain the precious space in the top menu purely as a means of helping you and others find the value you are looking for as easily as possible. So, I'm not changing the menu. Instead, I made an almost-just-as-easy way to support the site: chrismasterjohnphd.com/amazon. It is very easy to remember, you can type it right into your browser, and it goes [straight to the front page of Amazon](#). A small but meaningful portion of your purchases will be paid from Amazon to ChrisMasterJohnPhD.Com as revenue at no extra cost to you when you use that URL.

Comments and Update Log

This is a temporary post to help you easily see what has been updated and place it in its proper context. Later in January, I will create a permanent update log within the resource itself. Each round of updates will get a temporary new post like this one that will later redirect to the resource itself. Therefore, if you would like to leave a comment, please do so in the [comments section of the resource itself](#).

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